

Applied A.I. Solutions

Agile Project Management Methodologies

Quotations, Reference and Recommended Bibliography:

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Agile Project Management Methodologies

Ritual + Rhythm + Communication + Value +
Commitment + Teamwork + Discipline + Transparency
+ Continuous Improvement + Rapid Development

Week 2 / Day 2

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Agile Frameworks

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- An agile framework can be defined as a specific software-development approach based on the agile philosophy articulated in the Agile Manifesto.
- The Agile project management (APM) framework is an iterative and cohesive method of effective project management during the entire lifecycle of the project.
- Since 2001, many agile frameworks have emerged such as Scrum, Kanban, Lean, and Extreme Programming (XP).
- **Scrum** is the most popular agile framework, with 66 percent of all agile frameworks being scrum or scrum variants.

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Agile Frameworks

- Scrum
- Extreme Programming (XP)
- Dynamic Systems Development Method (DDSM)
- Feature Driven Development (FDD)
- Adaptive Software Development (ASD)
- The Crystal Method
- Lean Software Development (LSD)
- Disciplined Agile (DA)
- Scaled Agile Framework (SAFe)
- Rapid Application Development (RAD)

<https://www.productplan.com/glossary/agile-framework/>

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Which Framework is Best?

Factors that may influence which framework you choose:

- Company size
- Team structure
- Available resources
- Needs of stakeholders
- Structure/size of your product portfolio

<https://www.productplan.com/glossary/agile-framework/>

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Agile Project Management Framework Phases

1. Envision

- The Product is conceptualized, and all the project stakeholders are identified.
- The possible objectives of the project and the identification of the customer's needs are also a part of this phase.

<https://kissflow.com/project/agile/5-phases-of-the-apm-framework/>

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2. Speculate

- It deals with creating a features list of the final product and how the team would work to achieve it.
- The speculation phase normally revolves around two key activities:
 - Breaking down the project into a series of high-level milestones
 - Deciding the expected project timeline
 - Understanding of the critical tasks of the project
 - Prioritizing tasks
 - Deciding on quality of the final project deliverable

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3. Explore

- Explore the various alternatives to fulfill all the requirements of the project while staying within the given constraints.
- The focus is on creating value, maintaining quality of the deliverable.
- Teams work by focusing on a single milestone and iterate until perfection is achieved.
- This phase works parallelly with the Adapt phase because teams may have to change their plan and execution-style if a customer demands it or if the feedback is not as expected.

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4. Adapt

- The ability to adapt to different circumstances allows the team to be prepared for anything that gets thrown towards them.
- By constantly taking feedback from customers and ensuring that each aspect of the project is up to the end user's requirements, teams can significantly increase their efficiency and effectiveness.

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5. Close

- Teams ensure that the project gets completed in an orderly manner.
- The final deliverable is checked against the updated requirements of the customers
- Teams analyze their mistakes to avoid them in the future.

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Agile Planning

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Agile planning is a project planning method that estimates work using self-contained work units called iterations or sprints.

Sprints are brief periods of 1-4 weeks in which a team focuses on a small set of work items and aims to complete them.

The 3 Stages of Agile Teams

1. Survival
2. Learning
3. Self-Sufficient Agile Team

<https://www.4mation.com.au/blog/the-3-stages-of-agile-teams-which-stage-is-your-team-at/>

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1. Survival

How to progress to the next agile stage

- Observe your team as a whole and each individual during standups and planning meetings to determine what improvements are needed.
 - What personalities make up your team?
 - What impediments are they encountering?
 - Are their impediments to do with their interpersonal or technical skills?
 - Do your team members live your company values?
 - Are they meeting their short-term goals?

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1. Survival – cont'd

Your team will learn from experience, however, with proper planning and structure, this learning process can be sped up.

- Keep your team's pipeline organized
- Select people who will work the best together on that particular task
- Give your team members visibility on their current and future tasks
- Plan 3 month in advance to prevent over-scheduling your team
- Ensure your team are aware of your client's business goals and values
- Remove interruptions from your team's work and eliminate multitasking
- Have your team contribute to weekly status reports

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- Ensure your team are aware of your client's business goals and values
- Remove interruptions from your team's work and eliminate multitasking
- Have your team contribute to weekly status reports
- Provide Learning Time for Your Team

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2. Learning

Their teething problems are gone, and they have the time to learn and improve their work, processes and skillsets. Team leaders and project managers take the time to implement improvements to their team.

a. Progressing to the Next Agile Stage

- Team learns how to work with each other. It's important to give them enough time to learn each other's strengths and weaknesses
- Use DISC profiling tools (dominant, influencers, steady, compliant), to get a good mix of personalities to form a high-performance team

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2. Learning – cont'd

b. Upskilling your Team

- Upgrade their technical skillset
- Improve how they communicate with clients and teammates
- Enhance their attitude and language to ensure clear communication
- Provide motivation to upskill:
- Discuss their career aspirations, determine how they can upskill to achieve these

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2. Learning – cont'd

c. Becoming a Cross-Functional Team

- One of the goals of a highly agile team. How to achieve it:
 - Cross-train individuals to enable team members to help others who may be struggling
 - Expose team members to projects that they're not as familiar with, which increases the likelihood they'll develop new skills.
- Team Leaders and Project Manager's responsibilities begin to decrease as the team transitions into self-sufficiency

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3. Self-Sufficient Agile Team

c. Becoming a Cross-Functional Team

- Processes are clearly defined and followed
- Adaptive planning, incremental development, early delivery and continuous improvement become part of the day-to-day processes
Teams will have retrospective sessions to discuss what they've learnt
- Team leaders step back and observe the team and identify opportunities for improvement
- Team will be able to reliably deliver projects quicker and with less errors

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Agile Strategic planning process

- Strategy is defined by the enterprise.
- Decisions on how much to invest are motivated by value measurement as projects are completed and products are steadily released.
- Teams complete work that strategically makes sense for the company on an ongoing basis, and resources are allocated according to what is paying off as expected, and what isn't.
- Agile strategic planning takes the overhead out of constantly having to refactor capacity/resource management.

<https://www.atlassian.com/blog/jira-align/agile-strategic-planning>

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Agile Strategic planning process – cont'd

Allows to build, measure, learn, and strategically pivot if necessary.

- **Planning cadence: 3 months/quarterly**
The length of each iterative cycle is enough to create a long-term plan against initiatives that take six to 12 months to build out.
- **Scaled agile planning: 12-18 months**
Rolling annual planning process, comprising four to six program increments, and each program increment is planned one at a time.
- **Empowers portfolio managers to decide which projects they work on**

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End of Week 2 / Day 2

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